

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Bhuvaneshwari J(192572141)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

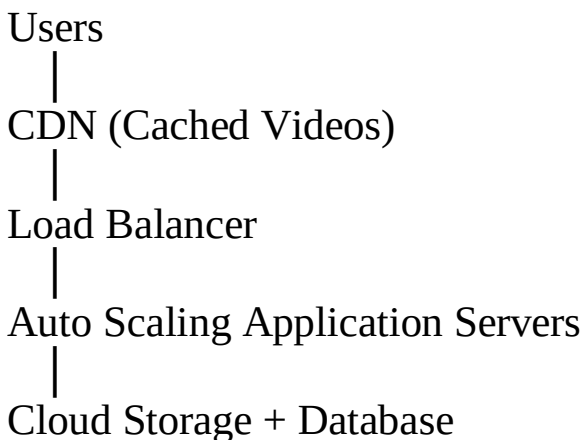
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. Media Company – Cloud Video Streaming Platform

a) Cloud Architecture

- Deploy the application on a cloud platform (AWS, Azure, or Google Cloud).
- Use **Content Delivery Network (CDN)** to deliver videos from the nearest server.
- Store videos in **Cloud Object Storage** (Amazon S3, Azure Blob Storage).
- Use **Load Balancer** to distribute user requests.
- Deploy application servers in **Auto Scaling Groups**.
- Use **Redis/Memcached** for caching frequently accessed content.
- Use managed databases with read replicas.

Architecture Flow



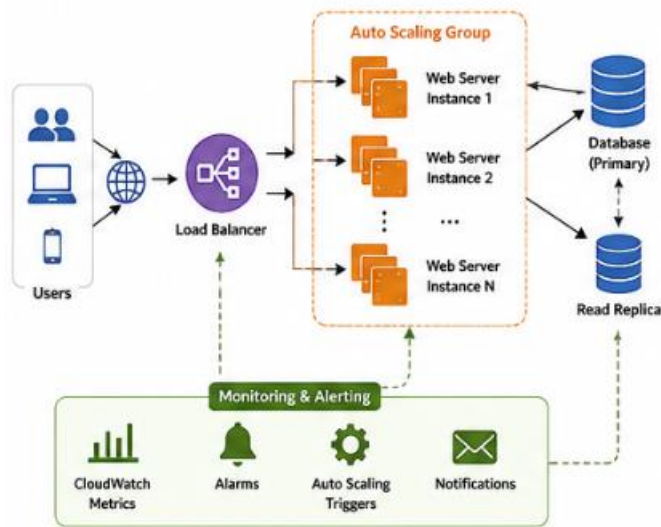
b) Load Balancing and Auto Scaling

- Load Balancer distributes traffic evenly among servers.
- Auto Scaling automatically adds servers during peak demand.
- Removes extra servers during low traffic.
- Prevents server overload and minimizes buffering.

c) Cost Optimization Strategies

- Use Auto Scaling to avoid paying for idle servers.
- Store old videos in low-cost archive storage.
- Use CDN caching to reduce bandwidth costs.
- Purchase Reserved Instances for predictable workloads.
- Monitor resource usage and remove unused resources.

1. Auto-Scaling and Load Balancing for Seasonal Sales



2. Fintech Startup – Multi-Cloud Containers

a) Solution Using Container Orchestration

- Package applications using **Docker** containers.
- Deploy containers using **Kubernetes**.
- Use Kubernetes clusters across AWS, Azure, and Google Cloud.
- Store container images in a container registry.

Architecture

Users

Global Load Balancer

Kubernetes Cluster (AWS)

Kubernetes Cluster (Azure)

Kubernetes Cluster (GCP)

b) Service Discovery and Scaling

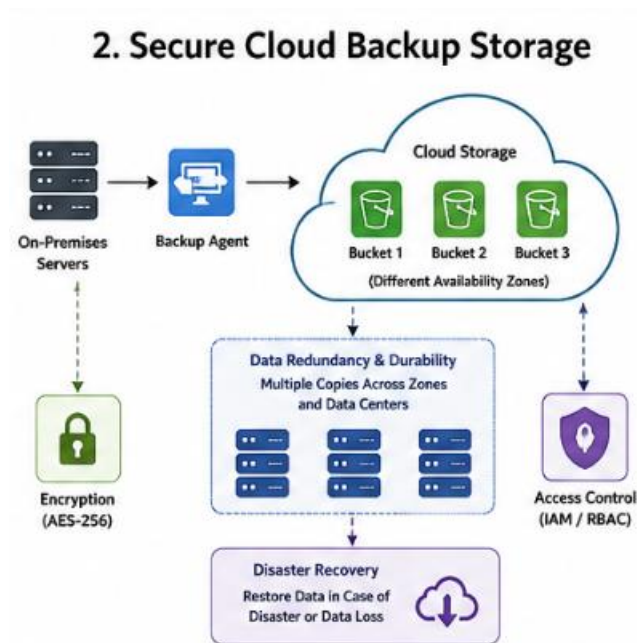
- Kubernetes Service Discovery automatically locates services.
- DNS resolves service names.
- Horizontal Pod Autoscaler increases or decreases containers.
- Load Balancer distributes requests across pods.

c) Benefits

- High Availability
- Vendor Independence
- Disaster Recovery
- Better Performance
- Scalability

Risks

- Higher management complexity
- Increased networking cost
- Data synchronization issues
- Security management across clouds

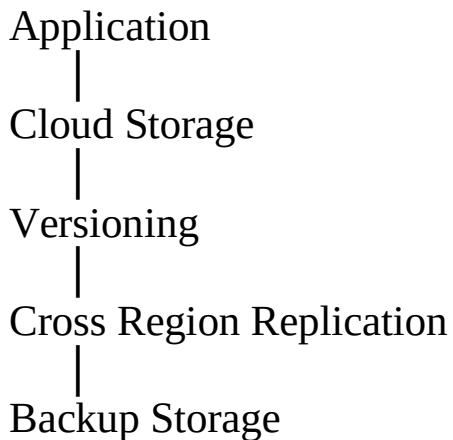


3. E-Learning Platform – Backup and Disaster Recovery

a) Cloud-Native Backup Solution

- Enable automatic cloud backups.
- Enable Object Versioning.
- Replicate data to another region.
- Schedule daily database snapshots.
- Store backups in multiple locations.

Architecture



b) Achieving RTO and RPO

Recovery Time Objective (RTO)

- Fast failover to backup servers.
- Automated recovery.
- Warm standby systems reduce downtime.

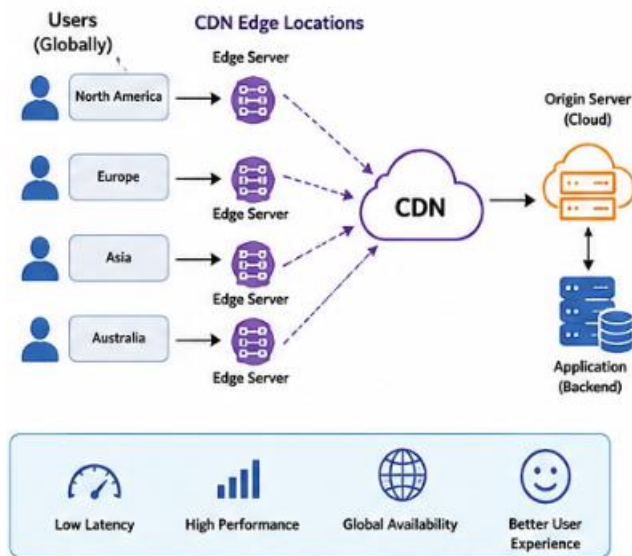
Recovery Point Objective (RPO)

- Frequent backups.
- Continuous data replication.
- Database transaction logs reduce data loss.

c) Business Continuity Steps

1. Perform automatic backups.
2. Enable cross-region replication.
3. Monitor backup status.
4. Test disaster recovery regularly.
5. Automate failover process.
6. Maintain disaster recovery documentation.

3. Global Deployment Using CDN



4. Government Agency – Hybrid Cloud

a) Hybrid Cloud Architecture

- Sensitive applications remain on-premises.
- Public applications run on public cloud.
- Connect environments using VPN or Direct Connect.
- Use API Gateway for secure communication.

Architecture

On-Premises Data Center



Secure VPN



Public Cloud



Applications

b) Identity Management and Encryption

Identity Management

- Single Sign-On (SSO)

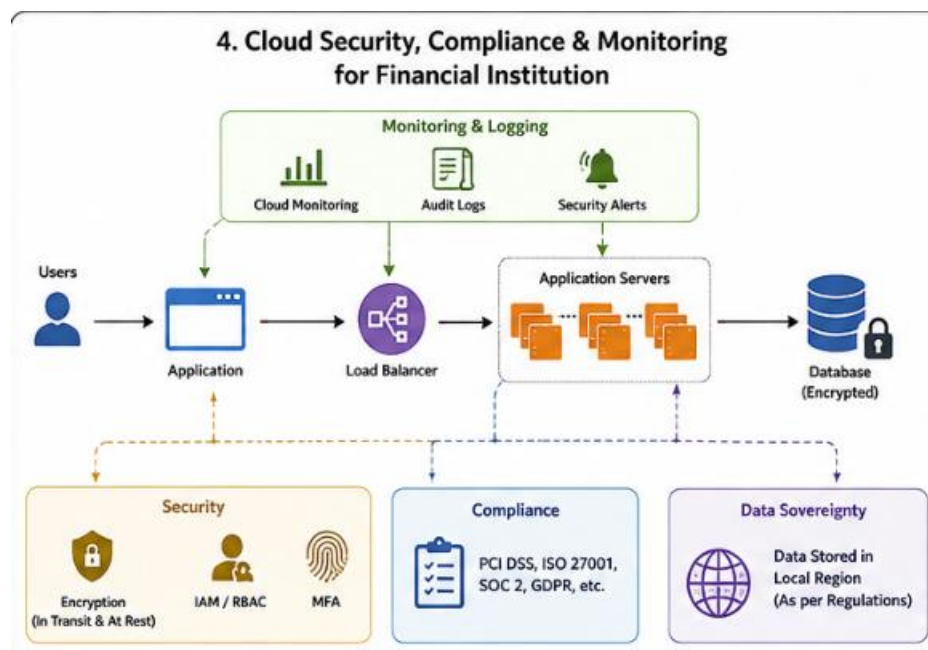
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)

Encryption

- Encrypt data at rest.
- Encrypt data in transit using TLS/SSL.
- Use cloud Key Management Service (KMS).

c) Challenges

- Complex management
- Compliance requirements
- Network latency
- Data synchronization
- Higher maintenance cost
- Security policy consistency



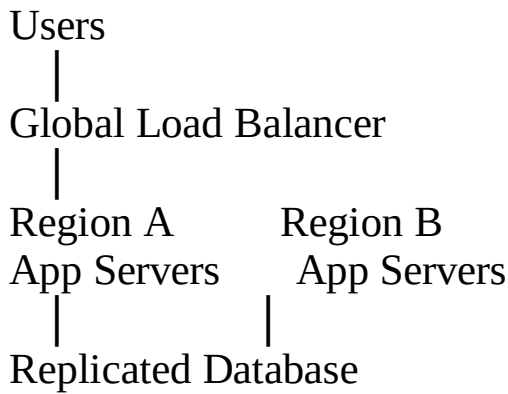
5. SaaS Provider – High Availability

a) Multi-Region Deployment Strategy

- Deploy applications in multiple cloud regions.

- Use Global Load Balancer.
- Keep identical application instances in each region.
- Use managed databases with replication.

Architecture



b) Data Replication and Failover

Data Replication

- Synchronous replication for critical data.
- Asynchronous replication for backup regions.
- Continuous database replication.

Failover

- Health checks monitor application status.
- Automatically redirect traffic if one region fails.
- Restore service with minimal downtime.

c) Monitoring and Alerting

- Monitor CPU, memory, and network usage.
- Enable application performance monitoring.
- Configure alerts for failures.
- Use centralized logging.
- Monitor database replication.
- Generate real-time dashboards.

5. Containerization and Kubernetes Orchestration

